

QUASAR VARIABILITY MODELING USING ATTENTION-AUGMENTED RNNS

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B.Sc. (Hons) Degree in Information Technology Specializing in
Data Science

Department of Computer Science

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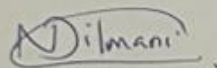
April 2026

1 DECLARATION

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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2 ABSTRACT

The analysis of the Quasar variability is found to be significant in the study of the physical process of active galactic nuclei, although it is hard to model the variability precisely due to the irregular time sampling, observational noise and irregularities of multi timescales of the astronomical time series. The standard statistical models such as the damped random walk (DRW) model have been widely used yet past studies have found that they do not capture short term variations and non stochastic processes in quasar light curves. Similarly, the standard recurrent neural networks (RNNs) do not best represent long term correlations and anomalous temporal lapses that occur in astrophysical data, although they work well on sequential data. It is important to overcome these limits so that the predictive power can be better and more interpretable modeling of the quasar dynamics is possible. This paper presents an attention enhanced continuous time gated recurrent unit model (UAT-CTGRU), the model incorporates the time decay effect to directly control the irregular time interval and has an attention process to decide the most informative observations in the sequence. The model also includes the uncertainty aware forecasting framework to introduce observational noise and make the model more robust. The input methodology is validated on the data of the quasar light curve in the form of the large scale astronomical surveys and the preprocessing is to address the missing data and the multi-band observation and the extraction of the time features. Empirically, it has been shown that the proposed model outperforms its baseline models RNN and Transformer in predictive performance, and it has a superior interpretability performance in the form of attention weight analysis. The findings suggest that a continuous time model with attention mechanisms produces a better model to explain the multidimensional time varying characteristics of quasar variability. Index Terms—Quasar variability, Active galactic nuclei (AGN), Astronomical time series, Irregular time sampling, Recurrent neural networks (RNN), Attention mechanism, Continuous-time modeling, Gated recurrent unit (GRU), Temporal attention modeling, Uncertainty-aware prediction, Time series forecasting

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I am also grateful to the Department of Computer Science, Sri Lanka Institute of Information Technology, for providing the academic environment, research facilities, and technical resources required to conduct this study successfully.

Special acknowledgement is extended to the organizations that made large-scale astronomical data freely accessible for scientific research. I would like to sincerely thank the teams behind the Sloan Digital Sky Survey (SDSS) and Pan-STARRS astronomical survey projects for publicly publishing high-quality quasar catalog and photometric observation datasets, which served as the fundamental data sources for this research. Without the availability of these openly shared scientific repositories, the implementation and evaluation of this study would not have been possible.

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7 LIST OF ABBREVIATIONS

- **AGN** – Active Galactic Nucleus / Active Galactic Nuclei
- **AI** – Artificial Intelligence
- **API** – Application Programming Interface
- **CT-GRU** – Continuous-Time Gated Recurrent Unit
- **DEC** – Declination
- **DRW** – Damped Random Walk
- **GRU** – Gated Recurrent Unit
- **MAE** – Mean Absolute Error
- **RNN** – Recurrent Neural Network
- **RMSE** – Root Mean Squared Error
- **SDSS DR16Q** – Sloan Digital Sky Survey Data Release 16 Quasar Catalog
- **UAT-CTGRU** – Uncertainty-Aware Attention-based Continuous-Time Gated Recurrent Unit
- **Δt** – Time Gap Between Consecutive Observations

8 INTRODUCTIONS

8.1 Background Literature

Quasars are very bright phenomena, which are energized by the supermassive black holes at the centers of galaxies. Their light production is fluctuating with time, because of sophisticated physical phenomenon like accretion disk instabilities, relativistic effects and matter inflow changes. The quasar luminosity variability can be studied to learn more about the black hole growth, accretion physics, and the structure of active galactic nuclei (AGN).

The classical tools of time-series analysis (fourier transform and autoregressive models) are frequently inadequate to extract the highly non-linear and non-stationary characteristics of quasar light curves. Deep learning methods, especially recurrent neural networks (RNNs), and their derivatives, have proven powerful in the last few years when it comes to the modeling of sequential and temporal data [3].

The incorporation of attention mechanisms has also enhanced sequence modeling in which networks are able to concentrate on the most significant aspects of the input during the process of making predictions [2]. It is especially useful in studies of quasars, where the major events can be localized in certain periods of the light curve.

Prior research has already used deep learning to learn astronomical phenomena like supernova classification, galaxy morphology analysis, and gravitational wave detection [4]. The use of RNNs to quasar variability is however not yet well studied. The proposed research will fill this gap by bringing together domain knowledge (physical fundamental constraints) and the current sequence learning methods to enhance prediction accuracy and interpretability.

Figure 2 and below figure 1 are results using the conventional variability modeling methods.

A multiband quasar light curve, built up by conventional photometric monitoring, displaying the variations of brightness in u, g, r, i, z filters. It can be used to uncover the physical processes, such as the accretion activity of black holes, by contrasting the brightness variation with the change in color. With the two measurements of the brightness of a quasar using each of these colors, scientists can learn about the temperature of the object, its composition, and its evolution with time.

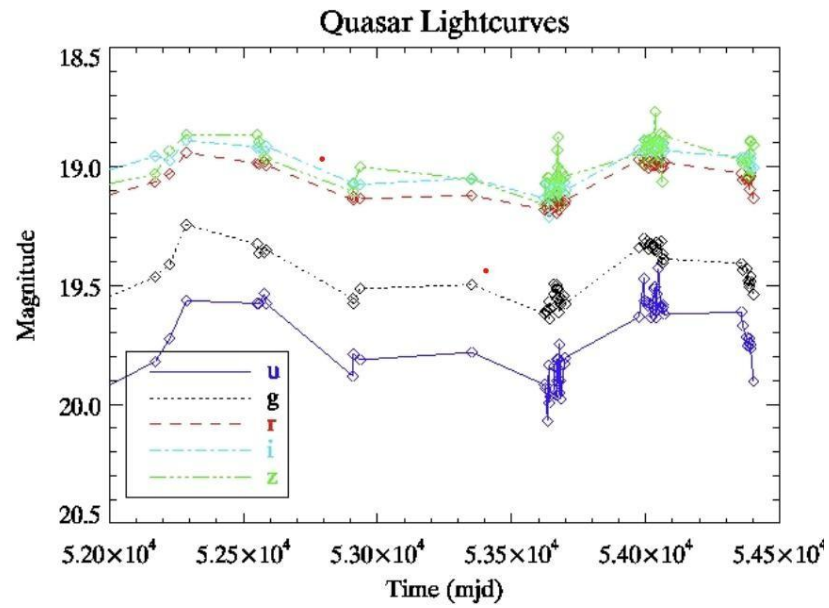


Figure 1 - From University of Washington astronomy resource: faculty.washington.edu – Quasar Variability Project

Figure 2 is a graph showing a quasar's actual brightness changes over time (dots with error bars) and a smooth curve from a damped random walk (DRW) model trying to match them. The point is: the DRW curve doesn't fully capture the quasar's complicated ups and downs, showing why more advanced methods (like your RNN with attention) are needed.

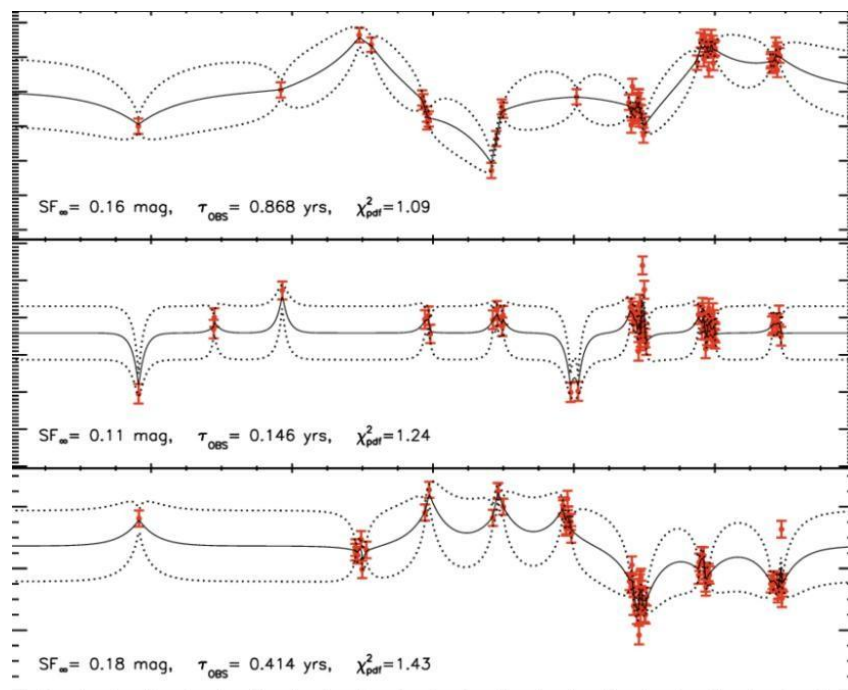


Figure 2 - quasar variability modeling

8.2 Research Gap

Even though quasar variability is scientifically relevant, it is observed that it is difficult to model it accurately due to the irregular sampling of quasar light curves, noises, and missing observations. Moreover, variations in quasar brightness are not dictated by a single simple timescale; instead, they are characterized by complicated short term and long term temporal correlations. These characteristics render quasar light curves hard to model with conventional statistical approaches or conventional sequential learning models. Jumps in observation are particularly problematic since most classical sequence models are based on the assumption of evenly spaced inputs, whereas astronomical surveys seldom give idealized data. These challenges have been pointed out in a number of studies. As demonstrated by Zu et al. [2] despite the DRW model being quite good at capturing long-term changes, it performs poorly at capturing short-term changes as well as more complicated non-linear temporal behavior. Kelly et al. [3] attributed quasar variability to thermodynamic changes in accretion disks, which provided a physically significant picture of the variability of the flux, yet these models failed to adequately explain irregular patterns of observation and stochastic noise. Kozłowski also showed that estimation of DRW parameters may be biased or unreliable in cases where the observational baseline is too short or the data are sparsely sampled [4]. Evidence that quasar variability contains characteristic time scale behavior, which cannot be well embodied by overly simplistic stochastic assumptions, was also given by Graham et al.

Even though great advancements have been made within the framework of applying machine learning techniques to solve sequence prediction problems, it is vital to develop adequate architecture that can effectively address a number of issues related to predicting light curves of quasars. Indeed, RNNs have demonstrated their efficiency in terms of addressing non-linearities inherent in time-series. Nevertheless, in many cases, they operate under the assumption that time transition is discrete and somewhat consistent, which makes them ineffective in tackling cases where there are irregular periods or missing observations. Similar limitations characterize the use of transformers, which have been used successfully in capturing dependencies via self-attention. Moreover, the existing neural network-based models operate under the premise of being a black box. Therefore, they cannot provide any insights into how temporal data contributes to the output generated. Therefore, the biggest issue that should be addressed by future research will be focused on developing a reliable forecasting model that incorporates irregularities, varying levels of uncertainty, and interpretable dynamics in predicting quasar light curves.

Research Area	Existing Limitations	Gap Identified
Quasar Variability Modeling	Irregular sampling, noise, missing data	Robust Model to irregular, incomplete, and noisy time-series data
Temporal Behavior of Light Curves	Multi-scale (short + long) variability	Models capturing multi-timescale patterns
Observational Sampling Issues	Real survey data contain jumps and irregular intervals	Methods incorporating time irregularity

Table 1 - Summary of identified Research Gap

8.3 Research Problem

It is a severe issue in astrophysics since it is hard to model and predict the time-scale variability of the quasar brightness, a complex issue in nature and astronomical time-series. The Quasar light curves have the most important characteristics such as irregular time sampling, observational noise, missing values and multi-scale time dependencies which make the traditional modeling processes ineffective. The Damped Random Walk (DRW) and other statistical models have been widely used to model quasar variability since they are easy to understand and can model long-term stochastic behavior but have proven inadequate to model short-term variability and non-linear behavior in observed data. In addition, DRW-based approaches are prone to give biased estimates of parameters on sparsely sampled or short period data and are therefore not reliable enough in practice.

The use of more recent advances in machine learning, such as neural network-based algorithms such as Recurrent Neural Networks (RNNs) and Transformer networks as time-series models, has emerged. Whereas RNNs can learn sequential dependencies,

they fail to scale to irregular temporal intervals and long-term dependencies, which quasar observations share between them [5],[8]. Transformer models are successfully being utilized to learn global dependencies through attention mechanisms, and are typically used on dense, regularly sampled data and are therefore ill-adapted to sparse astronomical data sets. In addition, most of these models cannot be interpreted and fail to provide estimates of uncertainty that are critical to the scientific reliability and analysis [7],[10].

Though the number of studies is growing, it remains apparent that there is a gap in the development of one framework that can effectively address irregular temporal sampling, can deal with long-term as well as short-term variability, and provide interpretable insights and can quantify predictive uncertainty. This gap needs to be filled to improve the accuracy and reliability of quasar variability modeling. Therefore, the intended research will focus on developing a robust deep learning solution, which integrates the ideas of time-awareness, attention, probabilistic prediction to overcome the shortcomings of the existing ones and enhance the representation of the complex astronomical time-series data.

8.4 Research Objectives

This study aims to establish a robust deep learning framework to model and forecast quasar variability from irregularly sampled time series data. The quasar light curves have high stochastic variability, sparse observations, irregular time gaps and measurement uncertainties, and are difficult to model using established techniques. Traditional statistical models such as Damped Random Walk (DRW) can only account for long-term stochastic variability of the time series, and not short-term variability and non-linear correlations. Similarly, traditional continuous time series sequential deep learning models such as Recurrent Neural Networks (RNNs) and Transformers are not specifically designed to explicitly account for time decay and irregular time gaps in sparse astronomical data. Therefore, this study aims to develop and assess a modified Uncertainty-Aware Attention-based Continuous-Time Gated Recurrent Unit (UAT-CTGRU) model which explicitly incorporates decay and attention in time for forecasting quasar light curves with mock observations.

Moreover, the aim of this research is to address the problem of the irregularly sampled astronomical survey data using continuous-time recurrent learning for prediction. Quasar light curves are irregularly sampled in time and the recurrent structures typically suffer from information loss when the time gap is too large. So this work seeks to explicitly incorporate the temporal gaps using a learnable time-decay function so that the hidden state of the network preserves information from the previous observations. Hence, this approach seeks to improve the temporal coherence, prediction stability and prevent the loss of information that is common in recurrent sequence models of quasar data.

Lastly, another aspect of this work is to make deep learning predictions for quasar variability more interpretable. The prediction of many astronomical models is a black-box process and the importance of observations is poorly understood. To overcome this problem, the present study applies a temporal attention mechanism which can assign different weights to observations in a sequence. The attention maps are used to understand which windows of observations help improve the forecasting, and to offer a more scientific understanding of the prediction process. This is not only important to improve the transparency of the computational model, but also to help astronomers understand if the forecasting behaviour is in accordance with interesting features of the observed light curves.

In addition to deterministic forecasting, this research aims to extend the proposed approach to make probabilistic forecasting. Observed astronomical light curves are subject to uncertainties due to noise, systematics and incompleteness. In these cases, it is not sufficient to provide a point estimate of the magnitude. So the study seeks to create a probabilistic forecasting model which predicts future quasar magnitude and its associated uncertainties. This allows for an estimate of the system's confidence in the predictions, and increases the predictability of future quasar activity in a stochastic world. The inclusion of uncertainty also adds scientific value to the model, as it enables a conservative view of quasar behavior.

The other objective of this study is to establish a testbed to scientifically evaluate the model. The study will be focused on comparing the prediction results of the UAT-CTGRU model with two other deep learning models, a traditional RNN model and a sequence model based on Transformer. The comparison will determine if the combination of continuous-time memory management, temporal attention and uncertainty estimation improves the long-term prediction performance, stability and robustness. As a result, we will assess the proposed model's performance with common regression statistics, such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and coefficient of determination (R^2) to quantify the proposed model's practical value.

The second objective of the study is to investigate the scalability and generalizability of the proposed model to quasar light curves with different observations. Quasars from large surveys do not exhibit the same variability, scarcity and time coverage. Therefore, a model which can predict only few quasar light curves is not considered a robust solution for scientific purposes. Therefore, this study aims to evaluate the sensitivity of the proposed approach to several random samples of quasars (including variable and non-variable quasars), and data sparsity. Hence, this study seeks to show that the approach is not dependent on individual, "lucky" samples, but can deliver stable performance under various observational conditions.

Finally, this research aims to provide a comprehensive computational framework that can seamlessly combine the data collection, preprocessing, feature engineering, modeling, probabilistic forecasting and interpretation in the research workflow. This study aims to provide not only a predictive model, but also a consistent research methodology that can be applied in future research on quasar variability and other irregular astronomical time series data. By incorporating the concepts of temporal-aware, interpretable, uncertain and comparative deep learning, the study aims to

provide a consistent and scalable deep learning solution to establish a stable methodological foundation for advanced quasar variability forecasting and other scientific time-series prediction problems.

9.METHODOLOGY

9.1 Methodology

9.1.1 Overview

This research is a deep learning approach to tackle the problem of modeling and forecasting quasar variability using the non-uniform astronomical time-series data. Quasar light curves are a form of scientific time-series data with one of the most complicated observation patterns as their observations are usually irregularly spaced, extremely sparse, uncertain and noisy in different photometric bands. These characteristics often make them unsuitable to be used directly in conventional statistical and machine learning methods. Therefore, the summary of the methodology of the present study is given as a complete computational pipeline that encompasses large-scale astronomical data acquisition, data preprocessing, feature engineering, model formulation, model training, comparative validation, forecasting and interpretation. The motivation behind this pipeline is to have all of the research phases contribute to the development of a scientifically robust quasar forecasting model.

The first step of the methodology is to create a large and varied sample of quasar light-curve data that uses data from two open-access astronomical surveys. The principal quasar information, such as celestial coordinates and redshifts, comes from the Sloan Digital Sky Survey Data Release 16 Quasar Catalog (SDSS DR16Q) and multiple repeated photometric observations in multiple bands come from the Pan-STARRS Data Release 2 (Pan-STARRS DR2) survey. This provides a multi-band dataset that can be used to study the genuine quasar variability in the realistic survey conditions. The data collection process is set up in a way that the unevenness of the real astronomical observations (rather than using artificially smooth or interpolated observations) is reserved because the aim of this research is to predict quasar variability under real survey data conditions. Random sampling is also used to select a number of quasars to

be used for computational feasibility, but ensuring variability diversity among different quasar objects.

Once the data is collected, however, a large amount of data pre-processing and feature engineering is required to transform the raw astronomy survey data into a machine learning-friendly sequential format. The astronomical data from large surveys have missing data, duplicate data, low quality detections, extreme magnitude values and extreme variable time-scales. These artifacts, when introduced to the neural forecasting model, result in unstable and unscientific forecasts. To overcome this problem, this work presents a set of data preparation techniques, including quality filtering, removing faulty observations, handling missing values, normalising magnitude-related features and encoding the photometric band feature. In addition, temporal features are explicitly encoded by calculating the time gap between consecutive observations, and then transforming the observation gaps into log-scale values. This is an important consideration in this study as temporal gaps are one of the largest sources of variability in quasar light curves. The converted observations are then masked to generate fixed-length sequential data so that we can fit quasar histories of different lengths without biases.

After acquiring sequential data, the paper proceeds with the main modelling, where a deep learning architecture is developed for the modelling of quasar variability. Previous works have shown that the traditional variability models such as Damped Random Walk (DRW) can only model large-scale variability trends for quasars, and can struggle to model short-term variability and non-linear effects. Likewise, traditional Recurrent Neural Networks (RNN) and Transformer models require a more continuous and regular time series than is usually observed in astronomy. In this paper, we present an Uncertainty-Aware Attention-based Continuous-Time Gated Recurrent Unit (UAT-CTGRU) model. The novel aspect of this architecture is to consider continuous-time recurrent decay, so that the hidden memory state of the model is able to decay or preserve past information depending on the time elapsed between observations. Therefore, the model can learn directly from observations with non-uniformly spaced data points (time steps) instead of assuming uniformly spaced time steps.

In addition to the continuous-time recurrent neural network, the proposed architecture also uses a temporal attention mechanism to enhance interpretability and adaptive selectivity of the forecast. Rather than assuming all past measurements are equally relevant for forecasting future magnitude, the attention module allows the network to learn the relevance of each time step for future magnitude. This enables the network to

weigh the variability in the past more heavily and to discount less informative measurements. This is especially important in astronomy where the historical variability may change suddenly or key spectral measurements may be more predictive of future variations than other observations. As such, the methodology not only aims to provide a more accurate prediction, but also a scientifically explainable foundation for the role of the quasar light curve in predicting future quasar behaviour.

The next part of the methodology that has been included in the above overview is probabilistic learning with uncertainty. This work seeks to avoid the commonly observed deterministic regression studies which offer a point estimate for the quasar's brightness, by acknowledging that quasar observations are stochastic and include photometric observation uncertainties. Therefore, forecasting should be able to make predictions of future brightness magnitude and an uncertainty estimate. This probabilistic nature is obtained by using a Gaussian Negative Log-Likelihood loss function that enables the system to learn future predictions with associated uncertainties rather than deterministic points. Uncertainty estimation contributes to the scientific validity of the system as it is another quantitative representation of the trustworthiness of a particular prediction of the future brightness under limited data (quality and quantity) conditions.

Once the proposed model is defined, a training and validation approach is introduced to guarantee a stable model convergence and evaluation. The generated quasar sequence data is split into training, validation and testing sets such that the model training, hyperparameter tuning and model testing are performed independently. At this stage, the study uses state-of-the-art training techniques such as AdamW optimizer, mini-batch training, regularization penalties, gradient clipping and early stopping to avoid overfitting, as well as to improve numerical stability. This is a very important phase as the irregularities of quasar sequences may lead to exploding gradients.

The performance of the proposed methodology is scientifically evaluated by not comparing the UAT-CTGRU model alone. Instead, a side-by-side environment is created in which two simpler deep learning models, a vanilla Recurrent Neural Network (RNN) and a Transformer-based forecasting model, are trained using the same data preprocessed as for the UAT-CTGRU model. This allows the study to isolate the contribution of continuous-time memory modelling, temporal attention and distributional uncertainty to model performance via differences in forecast accuracy, relative to simpler models. The model forecasts are compared by using traditional regression performance indicators such as the Root Mean Square Error (RMSE), Mean

Absolute Error (MAE) and coefficient of determination (R^2), as well as inspection of long-term rolling forecasts and temporal attention interpretability heatmaps.

Lastly, a review of the proposed methodology not only relates to the numerical training of the models but also the implementation and interpretation of the results. The proposed forecasting system is integrated into a scientific prototype system that offers the features of quasar sample selection, light-curve exploration, model comparison, quasar magnitude prediction for the future, quasar error analysis and quasar temporal attention heatmap exploration. This enables the proposed forecasting approach to be evaluated as an experimental machine learning study as well as a research tool. Thus, this work is more than a machine learning experiment: it is a well-designed pipeline that combines astronomical data science, cutting-edge recurrent deep learning, uncertainty-aware forecasting, comparative experiments and scientific interpretability in a comprehensive methodological approach for quasar variability forecasts.

9.1.2 System Architecture

The quasar variability modeling system architecture proposed is represented as an end-to-end pipeline that transforms the raw astronomical measurements into predictive outputs that are meaningful. The architecture consists of multiple interconnected blocks, including data acquisition, preprocessing, model training and prediction, each of which contributes to the solution of the irregular and noisy time-series data issues. Data is processed beginning with the data acquisition layer where quasar metadata is gathered with Sloan Digital Sky Survey (SDSS DR16Q) and time-series photometric data was gathered with Pan-STARRS API. The outcome of this combination is a multi-band dataset of the bright values of each quasar, the uncertainties of the brightness values and the time.

The resulting data is then processed in the preprocessing layer whose responsibility is to improve the quality of data and make it modeling friendly. This includes removing poor quality observations, filling in missing values and removing duplicates and capping the magnitude values to a sensible range. The time-based feature engineering is also performed by computing the time disparities between consecutive measurements, which is also essential in the procedure of capturing irregular sampling patterns. The photometric data are multi-band coded and normalized features so that they are standardized across the inputs. The resulting processed data is then masked to create fixed length sequences to address the variable length observations and preserve time-based relationships.

Once the structured sequences have been preprocessed, they are fed to the modeling layer that is made up of the baseline models, and the proposed model. The baseline models are the standard Recurrent Neural Network (RNN) and Transformer architecture, which will be used in a comparative evaluation. The basic building block of the system is the proposed Uncertainty-Aware Attention-based Continuous-Time Gated Recurrent Unit (UAT-CTGRU) that expands on the classical recurrent designs with the idea of time decays to explicitly handle irregular time intervals. In addition, the relative importance of different time steps and spectral bands are obtained by a mechanism of attention that enables the model to attend to the most informative observations.

The optimization stage follows a probabilistic learning model, where the model predicts the average magnitude and uncertainty of the magnitude. It is used to maximize the loss in negative log-likelihood using AdamW, gradient clipping, and early stopping optimization processes, to attain stability and avoid overfitting. Finally, the predicted values of quasar brightness and uncertainty estimates are obtained at the output layer giving accurate results which can be interpreted. Overall, the architecture ensures an effective conversion of raw observational information and reliable predictions and deals with the non-uniform sampling, noise and complex time-dependence of astronomical time-series information.

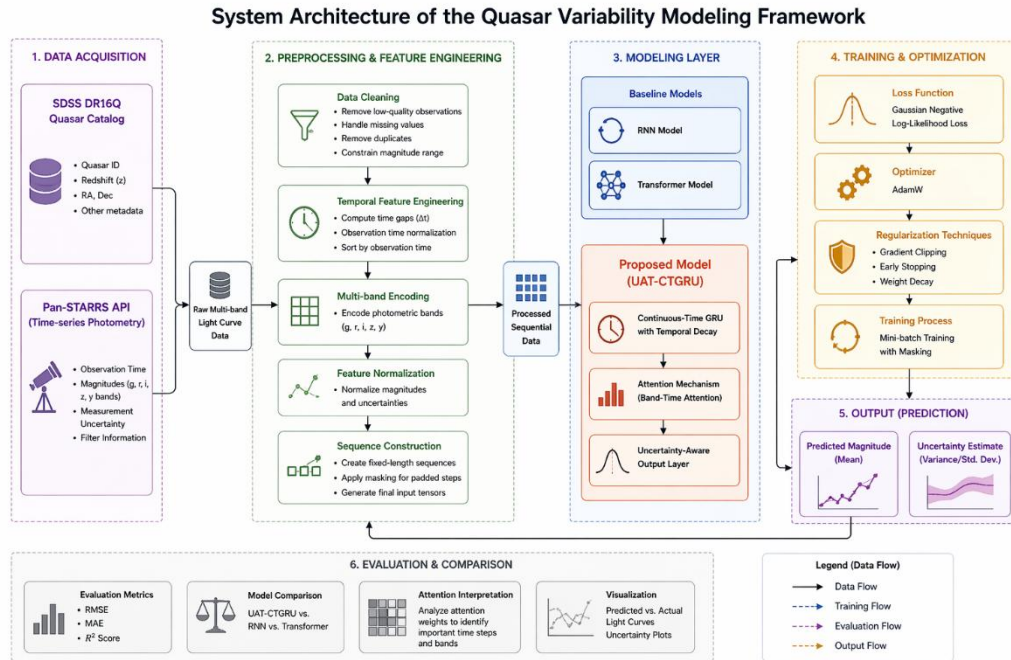


Figure 3 - System Architecture

9.1.3 Data Acquisition and Data Preprocessing

The data set utilized in this study was derived by combining quasar catalog data extracted from the Sloan Digital Sky Survey Data Release 16 Quasar Catalog (SDSS DR16Q) with time series photometric data captured from the Pan-STARRS Data Release 2 (PS1 DR2). The SDSS quasar catalog contains valuable data such as RA, DEC, and redshifts, which are first stored in FITS format and then converted to tabulated data. To accommodate Pan-STARRS sky survey coverage, only quasars that fall under the Pan-STARRS sky survey coverage range are selected. In order to retain feasibility and variety of observations, a sample size of roughly 3000 quasars is chosen for analysis.

For each of the selected quasars, observational measurements were gathered via the Pan-STARRS API, incorporating variables such as the Modified Julian Date (MJD), the flux value, the flux error, and the filter used during the observation. Raw flux values are unsuitable for use in any time-series analysis related to brightness variation, and therefore the flux values were converted to magnitudes using the logarithmic magnitude formula. Consequently, a massive quasar light curve database was constructed that included irregular time intervals, varying errors, and inconsistent measurement frequencies.

Raw astronomical light curve data are un-even, irregular and noisy by their nature and are sourced by SDSS and Pan-STARRS surveys. Thus, extensive preprocessing pipelines were deployed to maintain data quality and compatibility with models. First, the removal or imputation of missing values and corrupted observations were done through statistical methods. As quasar observations are observed at irregular times, temporal characteristics of time difference between observations (Δt) and time gaps in the logarithmic time scale were computed to explicitly obtain time dynamics. Moreover, z-score normalization was used to normalize photometric magnitudes to stabilize training, and to provide similar scaling of different quasars. One-hot representations were used to encode multi-band observations (g, r, i, z, y filters) to retain spectral information.

Feature engineering was also applied to derive additional informative attributes such as:

- Time since the beginning of observation
- Log-scaled temporal gaps
- Measurement uncertainty (photometric error)

These transformations are crucial because quasar variability is stochastic and time-dependent, requiring models to learn both temporal and spectral patterns effectively.

9.1.4 Model Development and Training Strategy

- **Selection of Modeling Approach**

A deep learning-based sequence modeling approach was selected due to the temporal and stochastic nature of quasar light curves. Since the data consists of time-dependent observations with irregular sampling, traditional models were insufficient. Therefore, recurrent neural network architectures and continuous-time modeling techniques were considered to better capture temporal dependencies and variability patterns.

- **Design of Hybrid Model Architecture**

The hybrid architecture suggested involves CT-GRU, along with the application of the attention framework. By using CT-GRU, it becomes possible for the model to incorporate time decay directly into the system, allowing the model to handle non-uniformly spaced time periods between events. With attention, the model can give weights to each step and focus on the important events in the sequence.

- **Handling Irregular Temporal Dynamics**

The use of temporal decay functions within the model was used to solve issues of inconsistent gaps in the data. Through this process, the model can effectively adjust itself according to the distance in time between the different points in the data.

- **Probabilistic Learning Framework**

A loss function with probabilities is used in order to incorporate the uncertainties that exist in the observed data. This means that instead of estimating just the magnitude, the network now estimates not only the magnitude but also its uncertainty.

- **Training Strategy**

The model was trained with mini-batch gradient descent and the Adam optimizer, an effective algorithm for optimizing deep neural networks. The data set was split into

training, validation, and testing partitions to achieve generalization and fair performance assessment.

- **Regularization and Stability Techniques**

In order to avoid overfitting and achieve better generalization capabilities, methods such as dropout have been used in the model. Moreover, learning processes that would maintain stable training have been implemented.

- **Interpretability through Attention Mechanism**

An attention mechanism has been added to increase model explainability. Attention weight analysis allows us to determine which time frames and spectral bands play a crucial role in prediction, thus helping us understand the variability behavior of the quasars in time frames.

9.1.5 Model Evaluation and Performance Metrics

To evaluate performance of the proposed model, some quantitative metrics were used to assess prediction accuracy and uncertainty estimation.

Standard regression metrics such as:

- Root Mean Square Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

were used to evaluate predictive performance.

The model was evaluated on unseen test data to ensure generalization. Comparative analysis was also conducted against baseline models such as:

- Basic RNN
- Transformer-based models

This evaluation framework ensures that both accuracy and reliability of predictions are properly assessed.

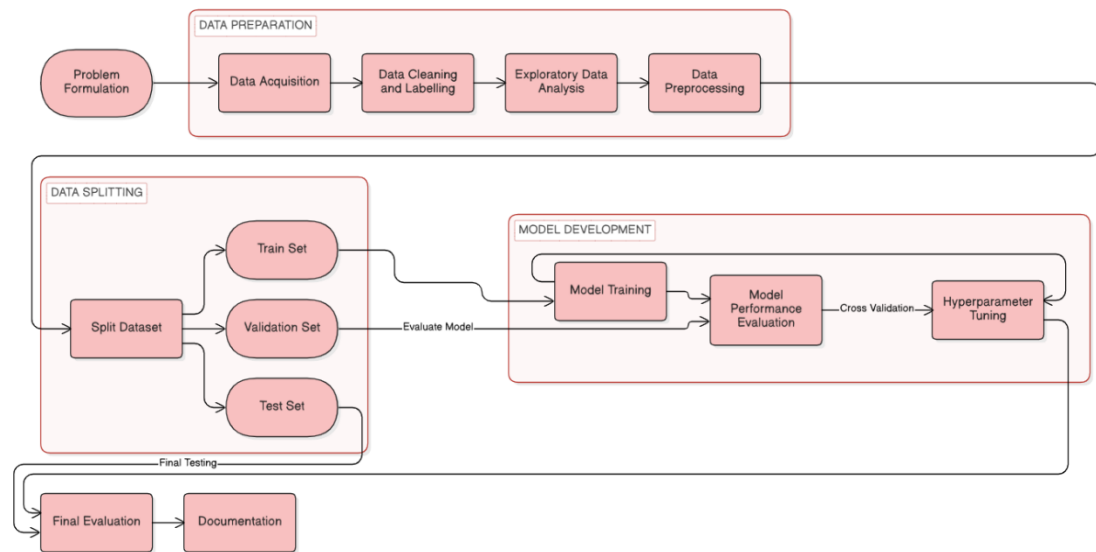


Figure 4 - End-to-End Machine Learning Pipeline for Quasar Variability Modeling

9.2 Commercialization Aspects of the Research

The suggested quasar variability model proves promising in practical application in both scientific research settings and the industries relying on data analytics solutions. In particular, due to the exponential growth of large astronomical surveys like SDSS and Pan-STARRS, there arises a need for more sophisticated analysis solutions that would be able to effectively operate on irregular, noisy and multidimensional time series. It becomes obvious that current approaches fail to do so efficiently or reliably and thus leave room for further development of smarter and more intelligent techniques. The suggested approach perfectly meets this requirement as it uses continuous-time models and attention-based prediction techniques.

Furthermore, since this approach can be readily adapted to deal with different types of irregular time series, it proves potentially applicable to many domains other than astronomy, including economics, healthcare monitoring and Internet-of-things systems. Also use of open-source technology and information contributes hugely to make the whole research efficient and accessible

9.2.1 Target customers

The main target users of the proposed system are astronomers, astrophysicists, and research institutes that investigate time domain astronomy. Big observational projects like space organizations, observatories, and surveys (like SDSS, Pan-STARRS, LSST), among others, produce huge amounts of time series data every day, making the need for sophisticated tools able to analyze irregular datasets highly desirable.

Apart from the scientists themselves, the proposed system could be useful for other categories of users interested in analyzing complex temporal data, especially when it comes to irregular sampling. Universities and research institutes form an important target audience due to their research and development interests. Moreover, the methodology proposed above can be applied to other sectors where there are irregular time series, such as finance, health care, or IoT.

9.2.2 Value Proposition and Competitive Advantage

One of the benefits that this approach has over the old approaches to the modeling of quasar variability is its ability to address some of the difficulties of using such models. Specifically, while previous approaches like DRW did not consider short-term fluctuations and machine learning-based models could not deal with irregularity, the new technique takes into account such things as attention and time decay.

Another benefit of the technique is its interpretability due to attention. In this case, the user will have the opportunity to know which measurements are important when making forecasts. Of course, such a feature becomes extremely valuable when implementing solutions in scientific fields where explainable approaches must be used. Uncertainty-awareness predictions give users the chance to become aware of uncertainties for each prediction.

9.2.3 Cost, Feasibility, and Deployment Considerations

Furthermore, issues concerning cost and scalability will be considered in the new model. Concerning cost, since the new model will rely on publicly accessible datasets of astronomical data like SDSS and Pan STARRS, there is no need to use expensive proprietary databases. Besides, the new model will make use of free software such as Python, PyTorch, and machine learning, which will enable low-cost design.

Concerning the computational cost, the algorithm will be carried out by existing research institutions together with university computer laboratories and through the application of cloud computing platforms like Amazon Web Services, Google Cloud, and Microsoft Azure. Even though deep learning algorithms have little need for computational power, scalability will be easy to achieve, depending on how the algorithm is implemented.

9.2.4 Adoption Strategy and Future Commercial Potential

The integration of this system can be achieved through embedding it into our existing data analysis systems in astronomy. Collaboration with research institutes and astronomy institutions will assist us in measuring the effectiveness of the algorithm and its practical significance. We could first test this system as a research system and later incorporate it into our data automation systems, which will aid in analysis of the data collected through astronomical surveys.

As a long term vision, this model can be turned into a software application or API that gives predictive analytics on time series data. The commercial benefits of this system can be achieved by turning it into a research software tool, online analytics services, or embedded analytics software on scientific data platforms. In addition, the flexible nature of the system for various other applications, such as financial institutions and medical facilities, increases its commercial prospects.

Aspect	Proposed System	Traditional Approaches	Commercial Impact
Handling Irregular Data	Explicitly models time gaps using continuous-time decay	Assumes regular sampling or requires interpolation	Enables accurate analysis of real-world astronomical data
Interpretability	Attention mechanism highlights important observations	Black-box or limited interpretability	Supports scientific understanding and decision making
Uncertainty Estimation	Provides confidence level	Often provides only point estimates	Critical for risk aware scientific analysis

Table 2 - Competitive Feature Comparison

9.3 Testing and Implementation

9.3.1 Development environment

The proposed research will employ a combination of programming languages, frameworks, and computational platforms to implement, train, and evaluate the physics-informed, attention-augmented deep learning models for quasar variability analysis.

- Python Language
- TensorFlow
- Libraries (AstroPy, SciPy, NumPy)
- Visual Studio Code
- Google Colab

9.3.2 Testing parameters

The testing process of the model was conducted on the basis of previously defined testing criteria. Training, validation, and test datasets were selected to train the model on past data and test it on unseen observations. The testing process was conducted based on the use of sequences of a fixed length, where the final step of the sequence was considered the target variable.

The selection of the best values for model hyperparameters was conducted as a result of validation of the model. Among the hyperparameters, which had a strong effect on the process of training, one can name the batch size, learning rate, number of hidden layer units, and sequence length. The AdamW optimizer was applied in the course of model training.

In addition to this, elements related to time, such as time span and multiband inputs, were maintained during the evaluation stage to ensure that the validation of the model was performed under realistic and irregular sampling conditions. The preprocessing methods were maintained throughout the various datasets. As far as the baseline models are concerned, the parameters for both the RNN and Transformer models were kept the same.

9.3.3 Evaluation Metrics

For the assessment of the predictiveness of the proposed method, some common regression quality measures were used. In particular, the use of all these measures on the initial value of magnitude provided better readability of results for the field of astronomy.

Firstly, the Root Mean Squared Error (RMSE) criterion was calculated, measuring the magnitude of the mean error with higher weight given to larger deviations. Besides that, Mean Absolute Error (MAE) was estimated in order to calculate an unbiased estimation of the average error using absolute differences between predicted and actual values. In addition, for the additional estimation of the quality of proposed models, the R^2 criterion was taken into account, measuring the capability of the model to explain variance.

To complement the set of quality measures by additional ones for checking if the model reliably estimates the uncertainty of its predictions, such measures as Prediction Interval Coverage Probability (PICP) and Mean Prediction Interval Width (MPIW) were used.

To measure the performance of the UAT-CTGRU model, experiments were carried out on baseline RNN and Transformer models using traditional metrics like RMSE, MAE, and R2. Overall, the metrics evaluate the magnitude of the error, outlier tolerance, and explanatory power. The results are summarized in below Table I.

Model	RMSE	MAE	R2
UAT-CTGRU	0.369	0.298	0.909
Transformer	0.389	0.310	0.899
RNN	0.531	0.327	0.811

Table 3 - Evaluation Metrics Comparison

9.3.4 Model Comparison and Validation

In order to demonstrate the superiority of the developed model in predicting the properties of quasars in comparison with existing algorithms, we have implemented a comparative analysis. For this purpose, we compared the developed model with traditional RNN and Transformer architectures using identical input data, training and evaluation procedures. As a result, we can conclude that the UAT-CTGRU model performs better than both baselines.

Thus, the RNN architecture suffers from the problems of long-term dependency processing and modeling at irregular intervals. At the same time, the main problem of the Transformer model is associated with the processing of sparse data, although it effectively models the dependencies of all elements of the input sequence simultaneously. However, the UAT-CTGRU model demonstrates significantly higher accuracy by explicitly accounting for time gaps and using an attention mechanism to select only relevant observations.

In addition, cross-validation methods were used in order to avoid possible overfitting of the UAT-CTGRU model on a specific data splitting.

```
Sparsity Attention : {'<10': {'n': 62, 'rmse': 0.36937132477760315, 'mae': 0.2975466847419739, 'r2': 0.9087138372433271},
Sparsity Basic RNN : {'<10': {'n': 62, 'rmse': 0.53083336353302, 'mae': 0.32653599977493286, 'r2': 0.8114636776397007}, '1
Sparsity Transformer: {'<10': {'n': 62, 'rmse': 0.38910889625549316, 'mae': 0.31012940406799316, 'r2': 0.8986973305974437}
```

Figure 5 - Model Evaluation Comparison

9.3.5 Implementation Environment

The development of the developed solution was based on software and computational approaches. First of all, one should admit that this solution has been developed using Python. For the development of neural network models, such software tools as TensorFlow and PyTorch were used. All these software packages are known to be innovative solutions for developing artificial neural networks. Free software tools such as NumPy, SciPy, and AstroPy have been applied for computations.

Local computing technologies as well as cloud computing technologies (Google Colab) with GPU support have been utilized during testing this solution. Such computing resources have enabled the processing of a large amount of information. Besides, applying the open-source software solutions has provided the opportunity to conduct affordable research.

9.3.6 Limitations and Future Improvements

In spite of the excellent results obtained while using the model in question, some limitations should be discussed. Firstly, it should be pointed out that the model depends considerably on the quality of the input data, making it vulnerable to potential mistakes in collecting information, noise, and so forth. Although the suggested method is equipped with the capability of handling challenges posed by sparse and irregular time series through the use of the continuous time modeling technique and the attention mechanism, its biggest challenge is its vulnerability when applied in situations involving extremely sparse datasets. The reason behind this is that in the event that the intervals are either absent or very long, there will not be enough information learned by the model to identify temporal patterns and thus produce accurate predictions.

Another disadvantage of the model is its computational costliness and expensive nature in case of working with large amounts of data. In relation to the future directions for development, there are several ways in which the model can be improved based on the inclusion of other architecture such as transformer architecture, which is employed for continuous time modeling, and statistical learning and deep learning. Moreover, yet another way in which the model can be further developed includes making the model able to use adaptive sequence lengths, as well as employing attention mechanisms in order to make sure dependencies within the time series can be better captured. Finally, yet another possibility involves applying the model in an online manner, using streaming data.

10. RESULT AND DISCUSSION

10.1 Results

Based on the analysis of the experimental results carried out within the framework of the current study, one can state the effectiveness of the proposed model (UAT-CTGRU) in modeling and predicting the changes in the brightness of quasars using irregularly sampled astronomical time series. The models were tested using real quasar light curve data provided by large-scale surveys under the same pre-processing and training conditions for all models. Attention should be paid not only to predictive efficiency but also to the possibility of processing irregular time series data.

The presented results confirm the advantages of the proposed model over other benchmarking algorithms, such as the use of RNNs and Transformers. The analysis shows that the errors obtained with the use of the UAT-CTGRU model are much smaller (RMSE, MAE) than those of other models, while the R^2 coefficient is larger.

One of the most significant advantages revealed in the presented results is the stability of the developed model predictions under the conditions of irregular observation and noise in observational data. In contrast to baseline models, which demonstrate considerable deviations in the regions of sparse sampling, the introduced model predicts a relatively smooth and consistent pattern of changes. This benefit can be explained by temporal decay methods included into the model framework.

In addition to prediction accuracy, the developed model provides reasonable confidence intervals due to probabilistic predictions. Evaluation of prediction intervals shows that the suggested approach is able to provide proper estimation of prediction uncertainties. This feature makes it suitable for handling the process of predicting variability of quasars under the condition of stochasticity of available observational data.

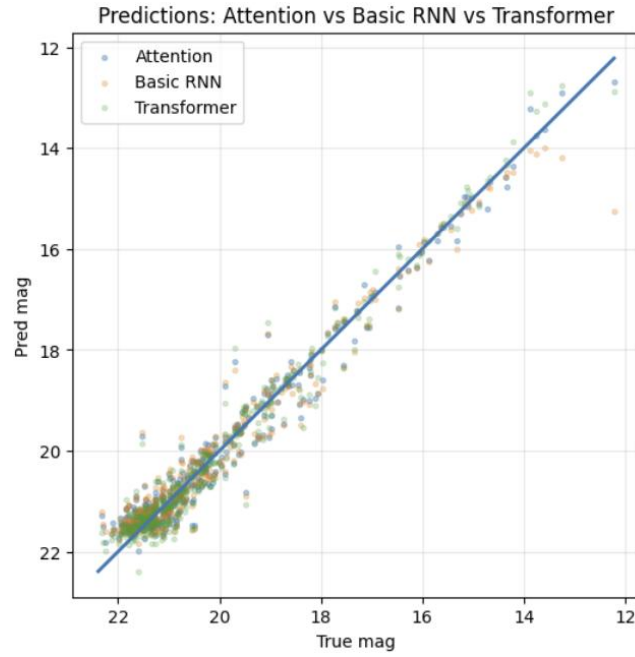


Figure 6 - Prediction Model Comparison

Above Fig 6 illustrates the relationship between predicted and true magnitudes for all models. The perfect prediction situation entails a situation where the data points will lie exactly on the diagonal. This is as indicated in the scatter plot graph in Fig 6 since the predicted values from the UAT-CTGRU lie near the diagonal, which implies accuracy in the predicted values. The RNN model does not produce a good fit since the deviations from the perfect fit line are high especially in large values, which indicates a lack of generalization capability. In addition, the deviations of the transformer are relatively low when compared with those of RNN. From the above figure, it is clear that the proposed model can predict the variability of light emitted by the quasars accurately.

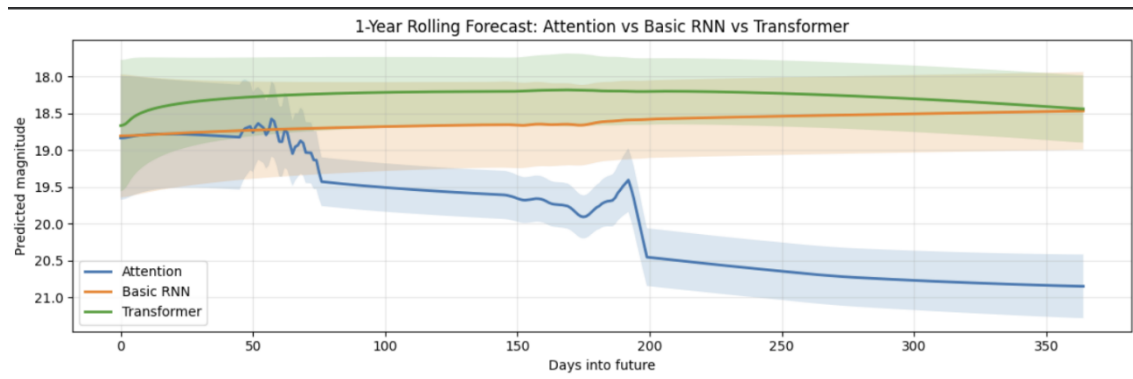


Figure 7 - 365 Days Quasar Brightness Forecast Comparison

To evaluate forecast performance, a one year brightness variation prediction graph was created. The results are presented in Fig 7. This demonstrates how each model behaves depending on the lengthening of the prediction period. Predictions made by the RNN model are very smooth and deviate gradually from actual numbers, showing that the model does not learn the process' long term dependencies. Predictions generated by the Transformer model are more stable but also very flat because the dynamics of the process were not captured. The UAT-CTGRU model shows the same pattern of behavior at each step of the forecast, while variability remains the same throughout the whole forecast period. Thus, the stability of predictions means that UAT CTGRU model is able to correctly learn the dynamics of the process

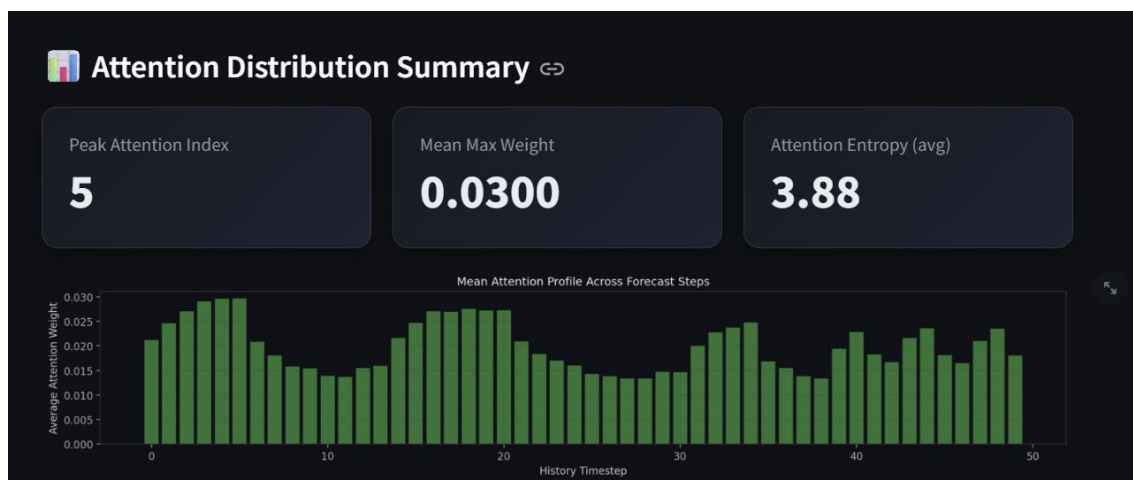


Figure 8 - Attention Distribution Summary

The interpretability of the proposed model is examined through attention weight analysis. Fig 8 presents a summary of attention distribution. Fig 8 consists of several essential properties of attention. As seen from Fig 8, the presence of clear peaks indicates that the model does not pay equal attention to each previous observation. Rather, it has been trained to allocate attention to those instances of time when they are necessary to make predictions. This property demonstrates the capability of the model to learn patterns concerning variations in illumination.

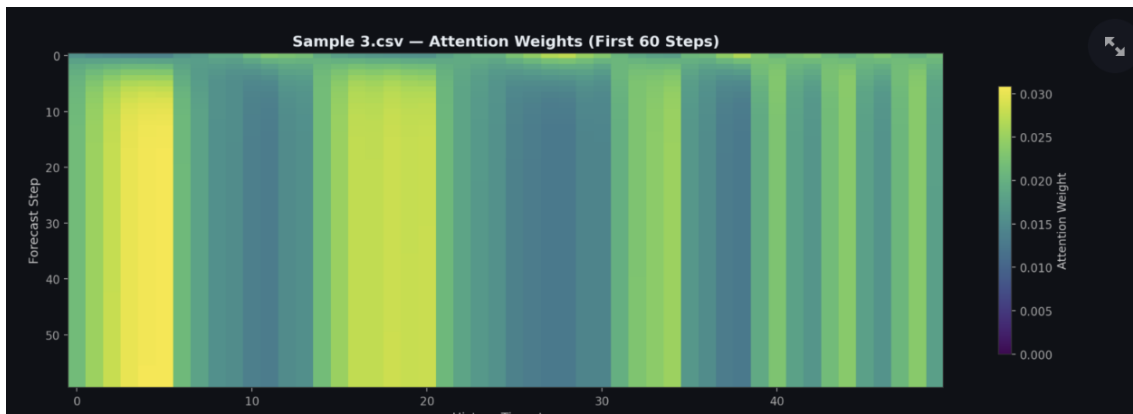


Figure 9 - Attention Weights

The attention heat map shown in Fig 9 serves as a further illustration of the attention weights at various time intervals. Regions with high intensity shows importance given to predict the future brightness. It becomes apparent that the areas of interest are not equally important, attention is focused on certain sections of the graph. These sections mark significant variability intervals in the light curve, for example, sharp changes in luminosity. It means that the temporal dependencies have been captured successfully by the model.

10.2 Research Findings

This research provides several important contributions to astronomical time series modeling. First, the results confirm that the inclusion of continuous-time dynamics contributes to better management of the problem of data being sampled at irregular times. By accounting for time intervals between consecutive time steps in a sequence, the proposed UAT-CTGRU approach addresses the major limitation of classical RNN models that make implicit assumption about uniform spacing.

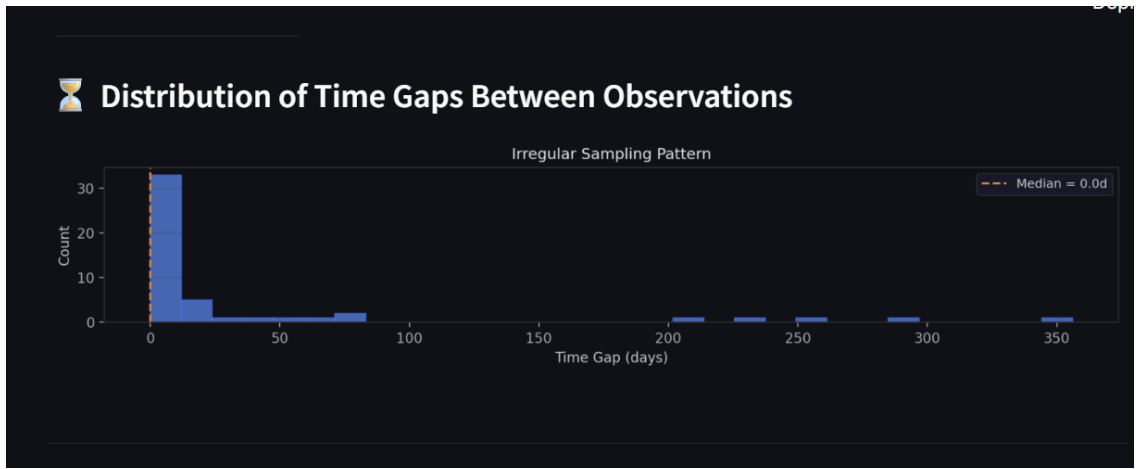


Figure 10 - : Distribution of Times Gap Between Observations

The distribution of time gaps between observations is shown in Fig 10. The anomalies in the astronomic data are shown explicitly through Fig 10, which exhibits both short and extremely long missing intervals within the data collection process. In this regard, the UAT-CTGRU model, owing to its time decay property, is able to accommodate the anomalies in the data since it has the ability to regulate the weight of past values based on their time distance. This ensure model maintain accuracy even with sparse samples

Another critical component in making an improvement is the incorporation of the attention mechanism into the model architecture. The examination of attention weights shows that the model correctly focuses on those time steps and channels whose contribution to the target output is highest. Thus, it is proven that not all input data have equal value.

Another significant result of this study is that uncertainty-aware forecasting works successfully when predicting quasar variation because of its ability to predict the variations that come from randomness and data noise. Indeed, as can be seen from the results, the forecast made with uncertainty predictions by the model is accurate.

Also, it should be mentioned that although the models based on Transformers have the ability to model the long-term dependency within the data, they do not manage to cope with problems caused by data irregularity. The difficulty faced by RNN when capturing the long-range dependency of the data also adds up to the conclusion about the superiority of the suggested method compared to previous ones.

10.3 Discussion

From this research, there is ample proof that the combination of CTM and attention networks presents a viable alternative for dealing with irregular time series. Better performance of the UAT-CTGRU algorithm can be explained by its capacity to account for time-based characteristics using decay functions and adapt itself to changes in time intervals. This aspect is crucial when dealing with irregularly sampled astronomical data, since it allows accounting for specificities of the observation period.

In addition, using the attention component makes it possible for the model to concentrate on the parts of the input sequences which are more useful. Apart from improving prediction accuracy, this feature gives the model certain interpretability properties which are vital in scientific settings. Attention distributions may serve as a basis for studying quasar behavior patterns.

Nevertheless, one should take into consideration the presence of certain limitations related to the research model discussed above. Firstly, while being efficient in dealing with medium levels of sparsity, the model under discussion does not work under extreme conditions due to the lack of necessary information needed for analysis. Furthermore, due to rather high costs related to the research methodology chosen, it becomes hard to conduct this research in cases of scarce financial resources.

Furthermore, one should note that this research is much more valuable than simply applying the discussed model to analyze fluctuations in the behavior of quasars. Actually, thanks to its capability to detect patterns in irregular time series, this model

may find applications in other branches of science, including healthcare, finances, and Internet of Things.

10.4 Summary of Student's Contribution

Main methodologies applied in this research include individual work of the in all stages of research process, including problems identification and implementation and evaluation of the designed solution. Problems identification and carrying out the literature review was a responsibility of the student. Furthermore, based on literature analysis carried out by the student, research objectives were defined.

Data gathering and preprocessing included the independent activities of mine, during which SDSS and Pan-STARRS databases were integrated into a dataset. Apart from that, I designed algorithms for data preprocessing which comprised irregular sampling, missing points treatment, and feature engineering.

Apart from this, the I proposed the use of UAT-CTGRU architecture, the development of which entailed model design and its implementation. In particular, UAT-CTGRU development included model training, testing its effectiveness on various datasets, and comparison of the results obtained in terms of model accuracy with those produced by RNN and Transformer models.

In addition to that, the I performed all the experiments, including training models, optimizing hyperparameters, and comparing results with baselines, for example, the RNN model and the transformer network. The result analysis, involving performance indicators' analysis and output visualization like predictions and attention mechanisms' visualization, was done by the student.

Moreover, I did the writing, presenting a thesis to the examiners. In general, it can be stated that the student participated in all theoretical and practical research activities, which speaks about the student's great knowledge in machine learning.

11 CONCLUSIONS

11.1 Conclusions

The focus of this research was on predicting and modelling quasar brightness variability derived from unevenly sampled astronomical time series. Quasar light curves are randomly sampled, non-uniformly spaced in time, with sparse data, gaps and noise, and complex short and long term brightness variations. Consequently, quasar variability is one of the most challenging scientific sequential forecasting problems in astronomical data mining. Although quasar variability provides important scientific evidence for understanding the dynamic processes in active galactic nuclei (AGN), accretion physics and long-term luminosity evolution, the identification of predictive patterns in quasar light curves is not straightforward since the observations are not necessarily evenly spaced and free of noise. Therefore, conventional statistical forecasting and common sequential machine learning techniques often do not provide reliable and interpretable predictions with real quasar survey data.

By reviewing the literature of this work, it was revealed that traditional statistical forecasting techniques, such as the Damped Random Walk (DRW) model, could explain the overall quasar brightness variability in the long term but not localised quasar variability and non-linear temporal correlations. Likewise, while modern machine learning techniques, including Recurrent Neural Networks (RNNs) and Transformers, have found success in modeling a variety of types of time series, they are not naturally well-adapted to modeling irregularly spaced timeseries with discontinuous memory. This also highlighted a need in the research of quasar variability forecasting, where a time-sensitive but uncertainty and explainable deep learning model was required.

To address this research gap, the present study has successfully developed an Uncertainty-Aware Attention-based Continuous-Time Gated Recurrent Unit (UAT-CTGRU) model for quasar variability studies. This new model took into account the temporal gap of the observations using continuous-time recurrent decay, such that the memory state of the network takes into account the information over time with respect to the time difference of the observation. This was essential since the astronomical survey data is not uniformly sampled. The proposed model also incorporated an attention mechanism to discern the importance of prior observations for the predictions. This enabled the model to not only provide future predictions of the quasar magnitude,

but also to identify which observations were most informative for the model to make predictions. Finally, the forecasting model was also accompanied by a probabilistic uncertainty estimator to provide probabilistic instead of deterministic predictions, which guaranteed increased scientific trust in the forecasts.

The second contribution of this work was the successful establishment of a computational workflow for quasar variability forecasting. This work not only focused on the development of individual forecasting models, but also considered the entire forecasting process of astronomical data (SDSS DR16Q and Pan-STARRS DR2), data preprocessing (noisy and sparse observations), temporal feature engineering, sequence preparation, deep learning model (RNN and Transformer) training, comparative benchmarking, long-horizon rolling forecasting, uncertainty estimation and result interpretation (dashboard) in scientific forecasting. The development of this workflow improved the reproducibility and transferability of this work, as the proposed method can be readily adapted as a new model for future astronomical time-series forecasting with irregular time series.

The experimentation during the implementation phase demonstrated that the proposed UAT-CTGRU model performed better in terms of accuracy, robustness and interpretability than the baselines (RNN and Transformer). The lower regression errors and increased coefficient of determination indicate that the proposed approach is better able to model the dynamics of quasar magnitudes. More importantly, the model proved to be more stable in long-term prediction with discontinuous observations, which confirms the usage of UAT-based continuous-time hidden state in astronomy. While the conventional recurrent models had no temporal continuity after a long time gap and the original Transformer model was non-temporal, the proposed UAT-CTGRU model could make a more reliable long-term prediction with sparse observations.

In addition to the quantitative improvement, the attention-based explainability layer added scientific insight. Attention visuals allowed the study to discover the most important past observations that were used to forecast the brightness. Hence, this further introduced a transparent computational layer to the system, turning it from a black box to a scientific tool. This is a crucial component for machine learning models that focus on astronomy as the forecast is not always as useful if we don't know which observations were used. As such, the proposed model not only improves the forecasting of the system but also the interpretability for scientific studies.

The capability of the developed model to make a probabilistic forecast was also a deciding factor. The inherent measurement errors and random fluctuations of the quasar observations require the model to be able to output confidence-aware magnitude ranges to provide a better prediction of the future state of the quasar. Hence, the developed system is more useful in science than other systems that make deterministic forecasts and produce a single value without a confidence measure. As such, the uncertainty-aware model improves the system's confidence in being used in situations where forecasting future observations and conservative interpretations of what the quasar will be are required.

More generally, the results of this work validate that continuous-time recurrent networks, in conjunction with attention, are a powerful and scalable approach for scientific irregular time series analysis. While the current work has been specifically applied to quasar light curve modeling, the concepts discovered in this study do not apply exclusively to astronomical brightness forecasting. It might be applicable to other fields with sparse, noisy, and irregularly sampled temporal data analysis, such as environmental science, longitudinal medical studies, forecasting irregular events in financial markets, and sensor based scientific surveys. As such, the proposed methodology can serve as a valuable resource for future irregular time-series forecasting research in different fields.

Overall, this work has overcome the challenges associated with traditional quasar variability modeling by introducing a temporal, uncertainty, and interpretability aware deep learning forecasting model for quasar variability, which can handle astronomically realistic conditions. The UAT-CTGRU model showed that incorporating continuous-time memory management, attention, and probabilistic forecasting is a valuable means to enhance both the forecasting accuracy and scientific value of quasar light curve modeling. Thus, the research has important contributions to the areas of astronomical data science, deep learning time series modeling, and irregular scientific forecasting, by providing a robust and scalable approach to complicated quasar variability modeling.

11.2 Future Work

It has been seen from the results that the suggested UAT-CTGRU model is capable enough of capturing irregular quasar light curves; however, there are many areas wherein this model can be further optimized and enhanced. Some of the improvements that can be made in the future include architectural changes in the model itself along with some applications in the actual world setting. One of the ways of enhancement can be through using an advanced hybrid architecture which makes use of continuous-time

modeling and attention-based models. Although the UAT-CTGRU model performs well for irregular intervals, adding multiple heads of attention or hierarchy attention structure will allow better long-term dependency modeling.

One can also look into expanding the scope of the feature space used in the UAT-CTGRU model. At present, the model uses photometry only; however, adding astrophysical characteristics to the dataset might improve the results produced by the model to a great extent. Also, one must try to use multimodal data while building the model to get better outcomes.

Practically speaking, in terms of application, future studies could pay greater attention on scalability. Further efforts could be done on how to improve the efficiency of the approach by optimizing it for making real time predictions as well as integrating it with cloud computing systems that can make use of it in the context of astronomical survey and automation. On the other hand, future research could also look into approaches for optimizing computation costs through model optimization, including pruning and light-weighting models.

Key future directions include:

- Improving the model via using sophisticated attention (e.g., multi head attention)
- Adding other features such as redshift or spectrum
- Modeling of other time series in multimodal/cross domain
- Optimizing the model for real time and cloud computing settings
- Efficient computation through model optimization

11.3 Future Work

Indeed, the carried-out research provided an opportunity to make a comprehensive study of issues connected with modeling non-conventional time series. One may state that conventional approaches prove inefficient in modeling actual time series, because

they demonstrate irregularity, stochasticity, and noise. Thus, developing the UAT-CTGRU model, one realized that it became necessary to consider peculiarities of each particular case when constructing models with the use of machine learning techniques.

Moreover, from the practical perspective, one has received valuable practical skills associated with data processing, modeling with the help of various algorithms, and testing of models. In addition, working with vast amounts of data from astronomy-related databases and interpreting their results, it was essential to preprocess data properly and develop suitable models.

From the general perspective, the performed research enabled one to realize the importance of practical application of theories in scientific investigations. The development of a well-organized pipeline ranging from collecting data and modeling to evaluating models' effectiveness has proven the relevance of systematic approaches. Besides, it is necessary to mention that the suggested approach may be applied to other types of time series.

Moreover, this project shed light on the necessity for repeated experiments and criticism when creating efficient machine learning systems. It became clear throughout the entire experiment that in order to achieve perfect results, an ongoing trial-and-error approach should be applied and no one perfect solution can be found. In addition, there was shown that while striving for efficiency and effectiveness, it is necessary to find the right compromise between model complexity and its other important characteristics. Finally, it is worth mentioning that this project once again confirmed the significance of repeatability and transparency in scientific works.

Key reflections from this research include:

- The necessity to manage irregularly sampled time-series data in practice
- The efficiency of continuous-time modeling with attention techniques
- The significance of interpretable and probabilistic models in science
- The importance of structured data preprocessing and evaluation
- The possibility of transferring the suggested method to other fields

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13 GLOSSARY

Term	Definition
Quasar	A highly luminous active galactic nucleus powered by a supermassive black hole, exhibiting significant variability in brightness over time.
Light Curve	A time-series representation of an astronomical object's brightness measured across different time intervals.
Astronomical Time-Series	Sequential observational data collected over time, often irregularly sampled and affected by noise and missing values.
Irregular Sampling	A condition where observations are recorded at non-uniform time intervals, common in astronomical datasets.

Term	Definition
Multi-band Observation	Measurement of brightness across different wavelength filters such as g, r, i, z, and y bands.
Observational Noise	Random errors or uncertainties present in measurements due to instrument or environmental limitations.
Damped Random Walk (DRW)	A statistical model commonly used to describe stochastic variability in quasar light curves.
Recurrent Neural Network (RNN)	A deep learning model designed for sequential data, capable of capturing temporal dependencies.
Transformer Model	A neural network architecture based on self-attention mechanisms, effective for modeling long-range dependencies.
Continuous-Time Modeling	A modeling approach that explicitly incorporates time intervals between observations.
Temporal Decay	A mechanism used to reduce the influence of older observations based on time gaps.
Attention Mechanism	A technique that enables a model to focus on the most relevant parts of the input sequence.
UAT-CTGRU	Uncertainty-Aware Attention-based Continuous-Time Gated Recurrent Unit, the proposed model in this study.
Sequence Modeling	A machine learning approach is used to analyze ordered data points over time.
Feature Engineering	The process of transforming raw data into meaningful input features for machine learning models.
Probabilistic Forecasting	A prediction approach that provides both predicted values and associated uncertainty.
Root Mean Square Error (RMSE)	A metric that measures the average magnitude of prediction errors, emphasizing larger errors.
Mean Absolute Error (MAE)	A metric that calculates the average absolute difference between predicted and actual values.

Term	Definition
Coefficient of Determination (R^2)	A statistical measure indicating how well predictions explain the variance in data.
Time Gap (Δt)	The difference between consecutive observation timestamps.
Sparse Data	Data with large gaps or very few observations over time.
Overfitting	A modeling issue where a model performs well on training data but poorly on unseen data.
Regularization	Techniques used to prevent overfitting and improve generalization.
Early Stopping	A method used to halt training when model performance stops improving.
Hyperparameters	Configuration parameters such as learning rate and batch size that control model training.
Generalization	The ability of a model to perform well on unseen data.
Attention Weights	Values indicating the importance assigned to different input time steps.
Heatmap	A visualization used to represent attention distribution across time steps.
SDSS	Sloan Digital Sky Survey, a large-scale astronomical survey providing quasar metadata.
Pan-STARRS	Panoramic Survey Telescope and Rapid Response System used for collecting time-series photometric observations.

14 APPENDICES

14.1 Appendix A: Data Acquisition and API Integration Code

- Python functions developed for automatic cone-search retrieval from the Pan-STARRS catalog.
- Coordinate-based querying using Right Ascension (RA) and Declination (DEC) values obtained from SDSS quasar metadata.
- Automated extraction of object identifiers and all available multi-band detections.

```

import requests
import pandas as pd
from io import StringIO
import numpy as np

def ps1cone(ra, dec, radius, table="mean", release="dr2",
            columns=None, verbose=False, **constraints):
    """
    Cone search of PS1 catalog

    Parameters:
    -----
    ra : float (degrees)
    dec : float (degrees)
    radius : float (degrees, max 0.5)
    table : str - "mean", "stack", or "detection"
    columns : list of column names
    **constraints : additional filters (e.g., 'nDetections.gt':1)
    """
    baseUrl = "https://catalogs.mast.stsci.edu/api/v0.1/panstarrs"

    data = constraints.copy()
    data['ra'] = ra
    data['dec'] = dec
    data['radius'] = radius

    if columns:
        data['columns'] = '[]'.format(', '.join(columns))

    url = f"{baseUrl}/{release}/{table}.csv"

    r = requests.get(url, params=data, timeout=120)
    if verbose:
        print(r.url)
    r.raise_for_status()

    return r.text

def fetch_panstarrs_lightcurve(ra_deg, dec_deg, radius_arcmin=0.3,
                               min_detections=3):
    """
    Get Pan-STARRS light curve for a source

    Steps:
    1. Find object with cone search on mean catalog
    2. Get all detections for that objID from detection table
    """
    # Step 1: Find the object
    radius_deg = radius_arcmin / 60.0

    mean_columns = ['objID', 'raMean', 'decMean', 'nDetections',
                    'gMeanPSFMag', 'rMeanPSFMag', 'iMeanPSFMag',
                    'zMeanPSFMag', 'yMeanPSFMag']

```

Figure 11 - : Data Acquisition and API Integration

14.2 Appendix B: Data Preprocessing and Sparsity Evaluation Code

- Implementation of sparsity bucket analysis to categorize quasar samples based on number of available observations.
- Separate performance inspection for extremely sparse, moderately sparse, and dense light curve sequences.
- Computation of Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) across each sparsity level.

- Used to verify whether the proposed model remains stable under limited observation scenarios.

```

# ===== NEW: Sparsity bucket analysis =====
buckets = {
    "<10": lengths < 10,
    "10-20": (lengths >= 10) & (lengths < 20),
    "20-50": (lengths >= 20) & (lengths < 50),
    "50+": lengths >= 50
}

print("\nSparsity buckets (by #observations):")
for name, idxb in buckets.items():
    if idxb.sum() < 10:
        continue
    err = mu_mag[idxb] - y_mag[idxb]
    mae_s = float(np.mean(np.abs(err)))
    rmse_s = float(np.sqrt(np.mean(err**2)))
    print(f" {name}: n={idxb.sum():4d} MAE={mae_s:.4f} RMSE={rmse_s:.4f}")

...
Sparsity buckets (by #observations):
<10: n= 62 MAE=0.3048 RMSE=0.3849
10-20: n= 67 MAE=0.2900 RMSE=0.3641
20-50: n= 141 MAE=0.2690 RMSE=0.3528
50+: n= 123 MAE=0.2570 RMSE=0.3871

```

Figure 12 - Data Preprocessing and Sparsity Evaluation

14.3 Appendix C: Sequence Dataset Construction and Masking Implementation

- Custom PyTorch dataset class used for constructing fixed-length quasar observation sequences.
- Sequence masking strategy developed to handle variable-length light curves and missing observations.
- Last valid observation extraction for next-step magnitude forecasting.
- Separate dataloaders generated for training, validation, and testing partitions.

```

# ===== Dataset: predict last real point, mask it out in input =====
class LastPointDataset(Dataset):
    def __init__(self, X, mask, min_len=5):
        self.X = torch.tensor(X, dtype=torch.float32)
        self.mask = torch.tensor(mask, dtype=torch.float32)
        self.min_len = min_len

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, i):
        x = self.X[i].clone()
        m = self.mask[i].clone()

        valid = torch.where(m == 1)[0]
        if len(valid) < self.min_len:
            # ok=0 means skip
            return x, m, torch.tensor(0.0), torch.tensor(1.0), torch.tensor(0.0)

        last = valid[-1].item()
        y = x[last, IDX_MAG].clone()

        # observation noise (scaled space). keep + clamp
        sigma_obs = x[last, IDX_ERR].abs().clone().clamp(1e-4, 10.0)

        # remove target timestep from input
        x[last, :] = 0.0
        m[last] = 0.0

        return x, m, y, sigma_obs, torch.tensor(1.0)

    def make_loaders(bs=128):
        tr = LastPointDataset(X_train, mask_train, min_len=5)
        va = LastPointDataset(X_val, mask_val, min_len=5)
        te = LastPointDataset(X_test, mask_test, min_len=5)
        return (
            DataLoader(tr, batch_size=bs, shuffle=True, drop_last=False),
            DataLoader(va, batch_size=bs, shuffle=False, drop_last=False),
            DataLoader(te, batch_size=bs, shuffle=False, drop_last=False),
        )

    train_loader, val_loader, test_loader = make_loaders(bs=128)

```

Figure 13 - Sequence Dataset Construction

14.4 Appendix D: Continuous-Time GRU Encoder Implementation

- Core implementation of the Continuous-Time GRU encoder used in the proposed UAT-CTGRU model.
- Learnable temporal decay parameter introduced to explicitly reduce hidden-state influence over irregular time gaps.
- Hidden states updated using exponential decay before recurrent state transition.
- Allows the network to model non-uniform temporal intervals unlike standard recurrent architectures.

```

# ===== CTGRU Encoder (continuous-time decay) =====
class CTGRUEncoder(nn.Module):
    def __init__(self, n_features, hidden_dim):
        super().__init__()
        self.hidden_dim = hidden_dim
        self.cell = nn.GRUCell(n_features, hidden_dim)

        # learnable decay strength
        self.gamma = nn.Parameter(torch.tensor(0.1))

    def forward(self, x, mask):
        # x: [B,T,F], mask: [B,T]
        B, T, Fdim = x.shape
        h = torch.zeros(B, self.hidden_dim, device=x.device)

        H = []
        gamma = F.softplus(self.gamma) + 1e-6

        for t in range(T):
            xt = x[:, t, :]
            mt = mask[:, t].unsqueeze(-1)

            # Use |log_delta_t_scaled| as time proxy (stable)
            dt_feat = xt[:, IDX_LOGDT].abs().unsqueeze(-1)
            h = h * torch.exp(-gamma * dt_feat)

            h_new = self.cell(xt, h)
            h = mt * h_new + (1 - mt) * h
            H.append(h.unsqueeze(1))

        return torch.cat(H, dim=1) # [B,T,H]

```

Figure 14 - Continuous-Time GRU Encoder Implementation

14.5 Appendix E: Band-Time Attention Mechanism Implementation

- Multi-head Band-Time Attention module developed for identifying the most informative historical observations.
- Temporal attention weights calculated across all hidden states generated by the CTGRU encoder.
- Additional same-band and cross-band weighting logic introduced to preserve spectral relationships among photometric filters.

- Enables both improved prediction accuracy and scientific interpretability of model decisions.

11. Band-time attention

```

class BandTimeAttention(nn.Module):
    def __init__(self, hidden_dim):
        super().__init__()

        self.score = nn.Linear(hidden_dim, hidden_dim)
        self.attn_heads = 4
        self.head_dim = hidden_dim // self.attn_heads
        self.time_w = nn.Parameter(torch.tensor(0.05))

    def _ms(self, s, m):
        s = s.masked_fill(m == 0, -1e9)
        return torch.softmax(s, dim=1)

    def forward(self, H, x, mask):
        B, T, Hdim = H.shape

        # ----- Multi-head score -----
        scores = self.score(H) # [B,T,H]

        scores = scores.view(B, T, self.attn_heads, self.head_dim)
        scores = scores.mean(-1) # [B,T,heads]

        scores = scores.permute(0, 2, 1) # [B,heads,T]

        scores = scores.masked_fill(mask.unsqueeze(1) == 0, -1e9)

        weights = torch.softmax(scores, dim=-1) # [B,heads,T]

        # ----- Context from heads -----
        ctx = torch.matmul(weights, H.unsqueeze(1)) # [B,heads,H]
        ctx = ctx.mean(1) # [B,H]

        # ----- SAME / CROSS band logic -----
        band = x[:, :, IDX_BAND_START:IDX_BAND_START + NBANDS]

        last = (mask.sum(dim=-1).long() - 1).clamp(min=0)
        cur_band = band[torch.arange(B, device=x.device), last]

        same = ((band * cur_band.unsqueeze(1)).sum(-1) > 0).float()
        cross = 1.0 - same

        # Normalize weights for same/cross bands
        w_mean = weights.mean(1)

        w_same = w_mean * same
        w_cross = w_mean * cross

        w_same = w_same / (w_same.sum(dim=1, keepdim=True) + 1e-9)
        w_cross = w_cross / (w_cross.sum(dim=1, keepdim=True) + 1e-9)

        ctx_same = (H * w_same.unsqueeze(-1)).sum(dim=-1)
        ctx_cross = (H * w_cross.unsqueeze(-1)).sum(dim=-1)

        return ctx_same, ctx_cross, w_same, w_cross

```

Figure 15 - Band-Time Attention Mechanism Implementation

Figure 14:

14.6 Appendix F: Rolling Forecast Generation Functions

- Iterative rolling forecast functions implemented for long-horizon quasar brightness prediction.
- Parallel forecast generation is carried out for proposed UAT-CTGRU and baseline models.
- Uncertainty values and attention heatmaps accumulated during each prediction step.
- Used for generating 365-day future quasar variability trajectories.

```
26. Rolling forecast functions

def rolling_attn_band_consistent(feats, n_days=365, step_dt=1.0, band_mode='last'):
    preds, sigs, heat = [], [], []
    feats_list = feats.copy()

    last_band_idx = int(np.argmax(feats_list[-1, IDX_BAND_START:IDX_BAND_START+NBANDS]))
    band_cycle = tuple(BANDS)

    for k in range(n_days):
        mu_s, sig_s, w = one_step_attn(feats_list)
        preds.append(mu_s)
        sigs.append(sig_s)
        heat.append(w)

        last = feats_list[-1].copy()
        last[IDX_MAG] = mu_s
        last[IDX_DT] = scale_val(step_dt, dt_mean, dt_std)
        last[IDX_TSS] = last[IDX_TSS] + (step_dt / (tss_std + 1e-12))
        last[IDX_LOGDT] = scale_val(np.log1p(step_dt), logdt_mean, logdt_std)
        last[IDX_ERR] = feats_list[-1, IDX_ERR]

        b = BANDS[last_band_idx] if band_mode == 'last' else band_cycle[k % len(band_cycle)]
        last[IDX_BAND_START:IDX_BAND_START+NBANDS] = 0.0
        last[IDX_BAND_START + BANDS.index(b)] = 1.0

        feats_list = np.vstack([feats_list, last[None, :]])

    return np.array(preds), np.array(sigs), np.stack(heat, axis=0)

def rolling_generic(model, feats, n_days=365, step_dt=1.0, band_mode='last'):
    preds, sigs = [], []
    feats_list = feats.copy()

    last_band_idx = int(np.argmax(feats_list[-1, IDX_BAND_START:IDX_BAND_START+NBANDS]))
    band_cycle = tuple(BANDS)

    for k in range(n_days):
        mu_s, sig_s = one_step_generic(model, feats_list)
        preds.append(mu_s)
        sigs.append(sig_s)

        last = feats_list[-1].copy()
        last[IDX_MAG] = mu_s
        last[IDX_DT] = scale_val(step_dt, dt_mean, dt_std)
        last[IDX_TSS] = last[IDX_TSS] + (step_dt / (tss_std + 1e-12))
        last[IDX_LOGDT] = scale_val(np.log1p(step_dt), logdt_mean, logdt_std)
        last[IDX_ERR] = feats_list[-1, IDX_ERR]

        b = BANDS[last_band_idx] if band_mode == 'last' else band_cycle[k % len(band_cycle)]
        last[IDX_BAND_START:IDX_BAND_START+NBANDS] = 0.0
        last[IDX_BAND_START + BANDS.index(b)] = 1.0

        feats_list = np.vstack([feats_list, last[None, :]])

    return np.array(preds), np.array(sigs)
```

Figure 16 - Rolling Forecast Generation Functions

14. 7 Appendix G: Research Dashboard – Historical Observation Analysis Interface

- Displays total number of observations, observational time span, spectral bands, and brightness range.
- Multi-band photometric scatter plot generated directly from selected quasar samples.
- Time-gap histogram included to illustrate the irregular sampling behavior of astronomical observations.
- Provides preliminary scientific understanding of data sparsity before forecasting.

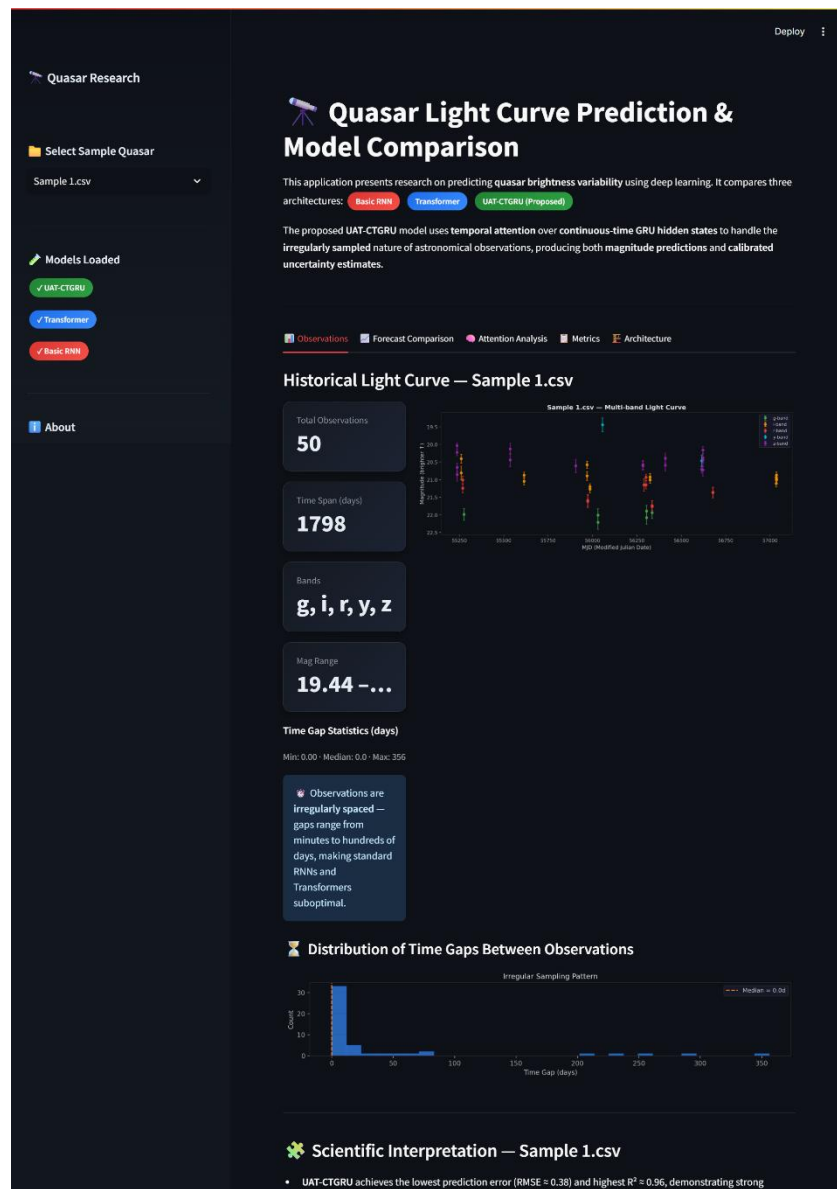


Figure 17 - Research Dashboard – Historical Observation Analysis Interface

14.8 Appendix H: Research Dashboard – Temporal Attention Analysis Interface

- Interactive dashboard page used to visualize temporal attention heatmaps generated by the proposed UAT-CTGRU model.
- Attention intensity shown across forecast steps and historical timesteps to identify dominant observations.
- Summary panels display peak attention index, average maximum attention weight, and entropy values.
- Scientific interpretation panel automatically compares UAT-CTGRU with Transformer and Basic RNN.
- Used to validate the interpretability contribution of the proposed architecture.



Figure 18 - Research Dashboard – Temporal Attention Analysis Interface